Weighted same-support: checked field bridge

Mathematical component report v3 — 2026-10-30

Result. The weighted field-polynomial theorem excludes the mixed zero-slope error branch of the Lean abstract classifier under the precise premises below. Producer checks and a fresh independent replay passed. Independent semantic review, preceded by a frozen statement-only derivation and controls, found the source faithful to this claim.

The checked declaration is `SameSupportCandidate.weighted_valid_inputs_exclude_same_support_error`.

Premises. Work over a field K, with decidable equality and $2 \neq 0$. For each $i \in \{1, 2\}$ require $z_i, w_i \neq 0$, and define

$$\bar{q}_i = q_i/z_i^2, \quad \bar{r}_i = r_i/z_i^4, \quad \bar{s}_i = s_i/(z_i^3 w_i), \quad \bar{t}_i = t_i/(z_i^5 w_i).$$

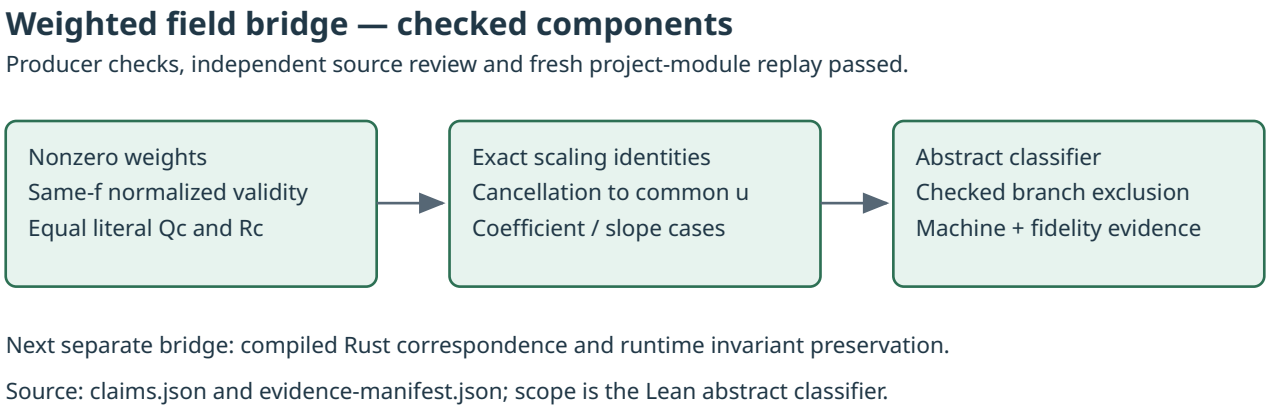
Both normalized monic quadratics $u_i = X^2 + \bar{q}_i X + \bar{r}_i$ divide $f - (\bar{s}_i X + \bar{t}_i)^2$ for the *same* polynomial f. Require $Q_{c1} = Q_{c2}$ and $R_{c1} = R_{c2}$, using the literal expressions

$$\begin{aligned} Q_{c1} &= q_1 z_2^2, & Q_{c2} &= q_2 z_1^2, \\ R_{c1} &= r_1 z_2^4, & R_{c2} &= r_2 z_1^4, \\ S_{c1} &= s_1 z_2^3 w_2, & S_{c2} &= s_2 z_1^3 w_1, \\ T_{c1} &= t_1 z_2^5 w_2, & T_{c2} &= t_2 z_1^5 w_1. \end{aligned}$$

Conclusion. The abstract classifier on $(S_{c1}, T_{c1}, S_{c2}, T_{c2})$ does not return `sameUMixedZeroSlope`. Equal S is an internal case split, not an additional premise. No desired coefficient-product identity is assumed.

Source identity. Published commit:
`957dd4c86b2a8b9416e4c62c6ad1f2eb51dcbee5`.
`WeightedSameSupport.lean` SHA-256:
`459d112522ace37aed602a51294fb9d30357114c47db1916d591ab5b0686e722`. Exact source and imported helper texts are retained in `evidence/producer`; machine and review bindings are in `claims.json` and `evidence-manifest.json`.

Proof connection and independent checks



With $Z = z_1 z_2$ and $W = w_1 w_2$, nonzero-weight cancellation gives

$$Q_{ci} = Z^2 \bar{q}_i, \quad R_{ci} = Z^4 \bar{r}_i, \quad S_{ci} = Z^3 W \bar{s}_i, \quad T_{ci} = Z^5 W \bar{t}_i.$$

The source proves normalized q, r equality directly from cross-product equality, and normalized slope equality within the equal-S case. It rewrites the second validity premise to the common support, then invokes the previously checked coefficient theorem. Its two products scale by $Z^8 W^2$ and $Z^{10} W^2$ to supply the classifier helper. The unequal-S case excludes the zero-slope branch directly. Inverse-before-double-before-zero-slope priority is material.

Independent replay. All five project modules were rebuilt from pinned sources in a fresh directory, followed by a separate declaration/axiom readback. All six processes exited zero. The inspected declarations report only `propext`, `Classical.choice`, and `Quot.sound`. Lean 4.34.1 used one thread, a 4,096 MiB limit and a 120-second external cap per compilation. The search path used the fresh directory plus the hashed dependency mirror, without producer project objects. Dependency builds were reused, not independently reproduced.

Semantic review. The reviewer froze a statement-only derivation and controls before receiving source, then froze the source review before compilation. The review covers the same-f divisibilities, all four nonzero premises, exact powers/orientation, cancellation, and branch order. A machine replay and a source-fidelity review are separately recorded evidence streams.

Next concrete bridge. Connect these literal field expressions and validity premises to the compiled Rust representation and operations, including invariant preservation at calls. This theorem concerns the manually defined Lean classifier; that implementation-refinement task is separate.

Boundary controls and operational record

These controls are exact hand derivations in the frozen blind review, not executed experimental trials. Their complete assignments are retained in `evidence/blind-review/BLIND.md`.

Control	Recorded observation
Valid unequal intercepts	Over $\mathbb{Q}$, weights one, $u = X^2$, $v_1 = 1$, $v_2 = -1$, $f = 1$: valid equal slopes and unequal intercepts are absorbed by the inverse branch.
Separate f_1, f_2	Inputs $(s_1, t_1, s_2, t_2) = (1, 0, 1, 1)$ can satisfy separate validity assertions but reach the excluded branch. Common-f validity prevents this.
Wrong S exponent	A recorded rational example changes exponent three to four, turning correct mixed slopes into equal slopes with unequal, noninverse intercepts.
Own instead of cross weight	A recorded rational example satisfies normalized validity but reaches the excluded branch when own weights replace opposite weights.
Zero weight allowed	Under Lean's totalized division, a zero-z_1 example satisfies normalized premises but has $S = (0, 0)$ and $T = (1, 0)$, reaching the excluded branch. Conventional partial division would leave the input undefined.
Characteristic two	No counterexample found by the stated algebraic argument: equal slopes imply equal intercepts through the square identity in characteristic two. The checked restriction is sufficient; necessity is not claimed.

Failures and warnings retained. An initial independent hashing driver failed before compiler execution because `hashlib.file_digest` was unavailable; its record and corrected driver are retained. The runtime helper emitted deprecated `if_pos`/`if_neg` warnings. The weighted source emitted unused/unreachable `ring` warnings. No compiler errors remained in the final independent replay. Exact command outcomes and durations are preserved in `outcomes.json` as operational records, without timing ratios.

Packet. This additive report preserves source, producer receipts, blind controls, source review, fresh replay outcomes, dependency/toolchain hashes and the failed-driver record. Generated Lean objects and runtime libraries are excluded. Prior reports remain unchanged. No research trial, cloud job or official ledger promotion accompanies report generation.